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9702/05

October/November 2007

1 hour 15 minutes

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

DO **NOT** WRITE IN ANY BARCODES.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

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1	
2	
Total	

This document consists of **8** printed pages.

- 1 Double glazing can be used for sound insulation. Double-glazed windows consist of two panes of glass with air in the space between them. Manufacturers reduce the air pressure in the space between the panes of glass to reduce the amplitude of sound transmitted through the window.

It is suggested that the amplitude A of sound transmitted through a double-glazed window is related to the air pressure p in the space between the panes by the equation

$$A = k\sqrt{p}$$

where k is a constant.

Fig. 1.1 shows a laboratory model of a double-glazed window. It consists of two panes of glass. There is a tube connected to the space between the two panes so that air may be removed.

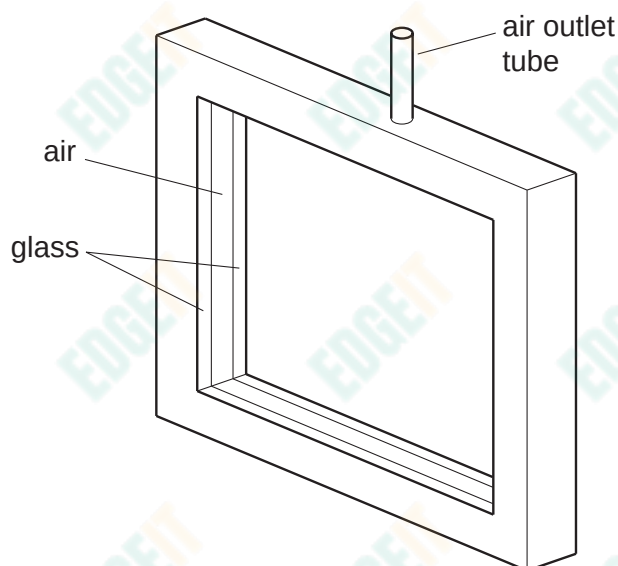


Fig. 1.1

Design a laboratory experiment to investigate whether A is related to p as indicated in the above equation when p is reduced. You should draw a diagram showing the arrangement of your equipment. In your account you should pay particular attention to

- (a) the procedure to be followed,
- (b) how the air pressure between the panes of glass would be measured,
- (c) how the amplitude of the sound would be measured,
- (d) the control of variables,
- (e) how the data would be analysed,
- (f) any safety precautions that you would take.

[15]

Diagram

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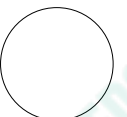
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- 2 An experiment is carried out to investigate how the current in a filament lamp varies with the potential difference across it.

An electric circuit is set up as shown in Fig. 2.1.

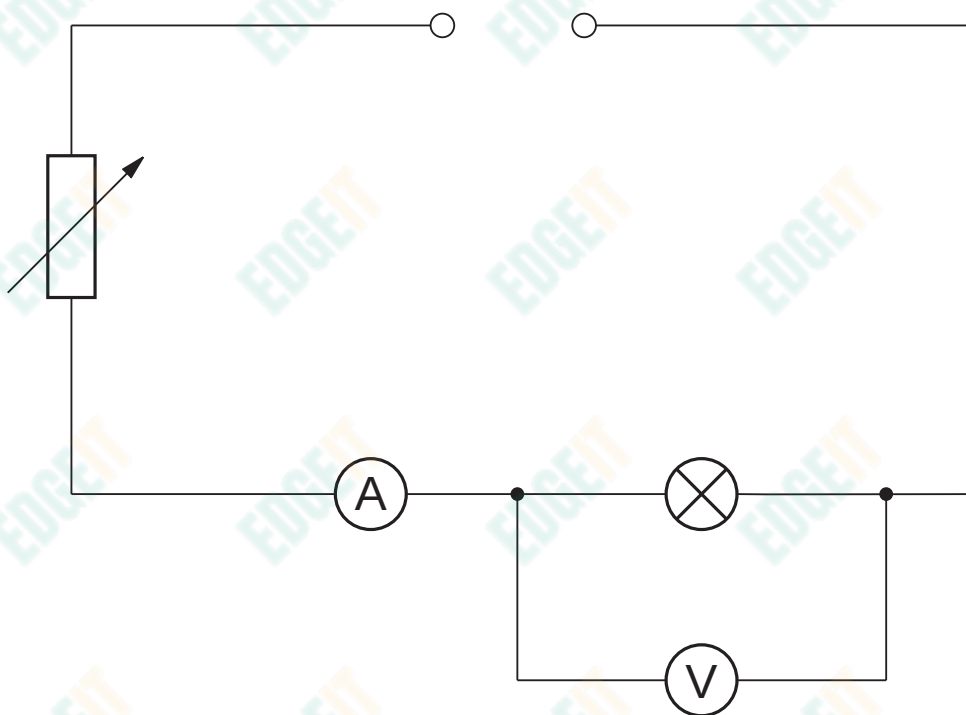


Fig. 2.1

Values of the potential difference V and the current I are given in Fig. 2.2.

V/V	I/A		
2.0	1.2 ± 0.1		
4.0	1.7 ± 0.1		
6.0	2.1 ± 0.1		
8.0	2.4 ± 0.1		
10.0	2.7 ± 0.1		
12.0	3.0 ± 0.1		

Fig. 2.2

It is suggested that V and I are related by the formula

$$I = kV^n$$

where k and n are constants.

- (a)** Explain why plotting a graph of $\lg(I/A)$ against $\lg(V/V)$ would enable you to confirm the relationship between V and I .

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 [1]

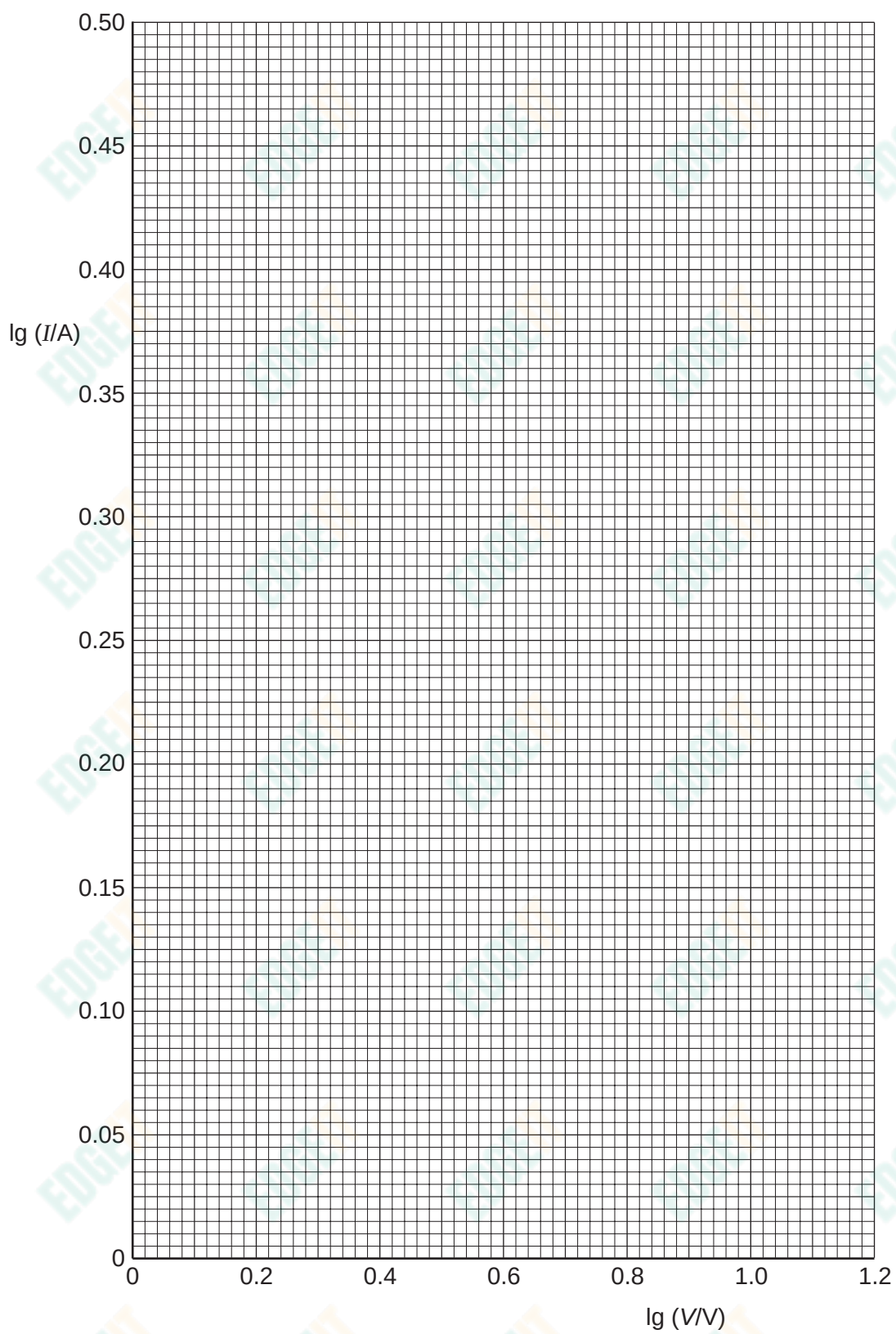
- (b)** Calculate and record values of $\lg(V/V)$ and $\lg(I/A)$ in Fig. 2.2. Include in the table the absolute errors in $\lg(I/A)$. [3]

- (c) (i)** Plot a graph of $\lg(I/A)$ (y -axis) against $\lg(V/V)$ (x -axis). Include error bars for $\lg(I/A)$. [2]

- (ii)** Draw the best-fit straight line and a worst acceptable straight line on your graph. Both lines should be clearly labelled. [2]

- (iii)** Determine the gradient of the best-fit line. Include the error in your answer.

gradient = [2]



- (iv) Determine the y -intercept of the best-fit line. Include the error in your answer.

y -intercept = [2]

- (d) Using the answers to (c) (iii) and (c) (iv), determine the values of k and n . Include the error in your values. You need not be concerned with the units of k and n .

k =

n = [3]

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